

NORTH OMAHA, NE, USA

WMO: 725530

Lat: 41.367N

Lon: 96.017W

Elev: 1332

StdP: 14.00

Time Zone: -6.00 (NAC)

Period: 86-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-0.1	-18.5	2.1	-5.1	-11.2	3.2	1.0	28.2	24.1	25.0	24.0	10.1	310	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	94.0	75.0	90.9	74.6	88.0	73.0	77.7	89.0	76.3	87.3	74.9	85.4	10.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.4	135.3	84.2	72.9	128.2	83.0	71.4	121.7	81.8	42.2	89.1	40.6	87.0	39.2	85.4	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	19.2	17.8	-11.4	97.9	6.5	4.8	-16.0	101.3	-19.8	104.2	-23.4	106.9	-28.1	110.4
			-12.2	80.7	6.2	1.4	-16.7	81.7	-20.3	82.5	-23.8	83.2	-28.4	84.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	27.3	28.5	41.0	52.2	63.8	73.0	75.7	73.3	65.6	52.4	37.5	27.5
	DBStd	20.49	11.41	12.86	11.98	10.68	7.94	6.90	6.06	6.74	8.33	9.92	10.73	12.05
	HDD50	2920	704	603	330	95	5	0	0	0	3	91	392	697
	HDD65	5981	1169	1023	747	410	118	14	3	11	94	403	827	1162
	CDD50	3507	0	1	52	160	432	691	796	724	471	165	15	0
	CDD65	1093	0	0	4	25	81	255	333	270	112	13	0	0
	CDH74	10052	0	0	47	278	668	2475	3165	2384	907	128	0	0
	CDH80	3808	0	0	10	94	169	987	1299	926	305	18	0	0
Wind	WSAvg	9.0	9.7	9.1	10.5	10.2	8.9	8.1	7.6	7.6	8.5	9.0	9.8	9.5
Precipitation	PrecAvg	30.3	0.6	0.6	2.0	3.0	4.1	4.0	3.8	3.3	3.6	2.5	1.3	0.9
	PrecMax	41.7	1.6	2.4	5.3	7.1	9.0	8.8	9.6	7.9	14.1	5.4	5.1	4.4
	PrecMin	21.3	0.0	0.0	0.0	0.4	0.6	0.0	0.0	0.6	0.0	0.0	0.0	0.0
	PrecStd	5.9	0.5	0.5	1.7	1.8	1.9	2.3	2.2	2.2	2.6	1.6	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	63.8	79.6	89.0	88.3	97.4	97.5	98.2	92.5	82.5	68.3	56.6
		MCWB	45.1	49.2	59.7	65.9	68.9	74.6	75.8	76.6	71.3	65.0	59.1	44.9
	2%	DB	52.2	55.5	71.9	82.1	84.6	92.6	93.6	92.4	87.7	77.8	62.1	49.4
		MCWB	41.9	44.6	55.1	62.4	67.7	74.1	76.0	75.9	70.7	61.9	54.3	41.0
	5%	DB	46.6	49.6	64.8	75.4	81.0	89.1	90.7	88.4	83.2	73.2	57.5	44.6
		MCWB	39.5	41.4	52.2	60.7	65.5	73.2	75.7	74.7	69.5	59.4	49.2	38.6
	10%	DB	42.1	44.6	60.0	69.9	77.6	85.7	87.3	85.2	79.2	68.3	53.6	40.6
		MCWB	36.4	38.3	51.0	56.9	63.8	70.9	74.7	72.7	67.8	56.8	47.2	35.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	49.6	61.6	67.9	73.5	77.9	80.0	79.6	75.4	67.1	61.4	45.7
		MCDB	55.0	62.6	72.5	84.3	83.5	90.4	90.8	90.9	84.6	80.1	66.1	53.6
	2%	WB	43.3	45.3	58.0	64.8	70.6	76.1	78.0	77.5	73.5	64.1	55.5	42.2
		MCDB	51.2	54.9	67.2	78.3	80.5	88.9	89.0	88.9	83.5	73.7	60.9	48.6
	5%	WB	39.7	41.8	54.5	61.9	68.1	74.2	76.6	75.9	71.4	61.2	50.7	39.3
		MCDB	46.2	49.3	64.8	73.1	77.6	86.6	87.5	86.2	80.6	71.0	56.5	44.5
	10%	WB	36.7	38.7	50.8	58.1	65.8	72.4	75.1	74.4	69.5	57.9	46.5	36.2
		MCDB	41.9	44.1	59.9	69.4	74.3	83.8	85.7	83.6	77.5	67.4	53.3	39.8
Mean Daily Temperature Range	5% DB	MDBR	18.1	17.2	20.0	21.1	19.4	18.9	17.5	17.8	19.8	20.4	16.6	15.4
		MCDBR	24.4	25.6	28.7	26.5	23.0	21.6	20.8	20.9	22.9	26.9	23.1	19.8
	5% WB	MCWBR	16.6	16.7	16.5	13.2	11.9	10.4	9.8	9.4	10.8	15.0	16.3	14.3
		MCWBR	23.0	24.7	25.7	25.1	20.4	20.4	19.3	19.4	20.3	25.2	20.5	19.1
Clear-Sky Solar Irradiance	taub		0.279	0.312	0.328	0.384	0.407	0.412	0.423	0.412	0.373	0.329	0.295	0.282
		taud	2.449	2.378	2.399	2.257	2.261	2.306	2.270	2.306	2.373	2.484	2.505	2.457
	Ebn at Noon		280	285	292	281	276	274	270	269	273	276	272	269
		Edh at Noon	24	30	34	42	43	41	42	39	34	27	23	22
All-Sky Solar Radiation	RadAvg		632	933	1207	1547	1837	2042	2060	1775	1464	1022	711	532
	RadStd		58	55	109	111	138	159	134	118	95	120	49	53

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page